

SHEREIFF GARRETT

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EDUCATION

Fayetteville State University

Bachelor of Science

Major in Computer Science; Minor in Physics

Cumulative GPA: 3.69/Major GPA: 3.72

Fayetteville, NC

January 2023-May 2026

PUBLICATIONS & PRE-PRINTS

Garrett, S. (2025). Machine Learning and Machine Learned Predictions in Chest X-ray Images. arXiv:2507.23455(Publication)

RESEARCH EXPERIENCE

FAYETTEVILLE STATE UNIVERSITY (Supervisor: Dr. Chandra Adhikari)

Materials Science Researcher

Fayetteville, NC

Sep 2024 – Present

- Conduct computational modeling of Ti_4N_3 MXene using VASP to analyze band structure, magnetism, and spin polarization
- Prepare and analyze CIF structures for density functional theory (DFT) calculations.
- Develop algorithms for graphs and data analysis using Python techniques and MATLAB.
- Research the correlation between nitrogen termination and net magnetization for 2D energy materials

NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION(NAWCAD)

Artificial Intelligence Research Intern (DoD HBCU/MI Research Internship)

Patuxent River, MD

May 2024 – Aug 2025

- Conducted Artificial Intelligence research on reinforcement learning for multi-agent systems in simulated environments, focused on autonomous swarm behaviors.
- Developed and evaluated agents such as Actor-Critic and Rainbow DQN using PyTorch and TorchRL.
- Implemented machine learning techniques to improve the efficiency and adaptability of autonomous systems.
- Developed and optimized a reward framework that improved swarm cooperation by 20%.

NASA L'SPACE (Arizona State University)

Command & Data Handling Engineer

Remote

Aug 2024 – Dec 2024

- Led the design of a spacecraft Command & Data Handling (CDH) subsystem to meet mission science objectives.
- Performed trade studies considering mass, power, cost, and schedule constraints.
- Research heritage systems to identify reusable subsystems for improved reliability.
- Developed and implemented CDH subsystem requirements for mission concept proposals.

AIR FORCE RESEARCH LABORATORY (AFRL)

Artificial Intelligence Research Intern (DoD HBCU/MI Research Internship)

Rome, NY

May 2024 – Aug 2024

- Developed algorithms that optimized ResNet-18 for Synthetic Aperture Radar (SAR) image classification to support Automatic Target Recognition (ATR).
- Developed and evaluated different metrics for accuracy and computational efficiency using PyTorch Lightning.
- Achieved 14% higher classification accuracy with noise and 35% faster inference.
- Researched transfer-learning techniques to handle limited imagery data.

INDEPENDENT PROJECTS

PNEUMONIA X-RAY CLASSIFICATION (ongoing)

2025

- Developed and implemented deep learning models (DenseNet-121 vs baseline CNN) for pneumonia detection in chest X-ray imaging, analyzing performance and interpretability. Using PyTorch libraries and Gradient-weighted Class Activation Mapping (Grad-CAM).

MATH SOLVER (ongoing)

2025

- Developing a real-time recognition system for handwritten digits and math symbols using YOLOv, training a model, and applying computer vision techniques to detect and parse equations for automated solving.

DECK VISION

2025

- Developed a PyTorch-based model built from scratch to detect and classify playing cards, trained on a Kaggle dataset to identify both suits and ranks with high accuracy. Designed the project to be a customizable framework for real-time object recognition tasks.

HEAT DIFFUSION PARALLEL SIMULATOR

2025

- Built a high-performance heat diffusion simulator in C++ using OpenMP, MPI, and CUDA, optimizing a 2D heat equation solver for parallel execution across CPU and GPU architectures. Implemented numerical methods, boundary conditions, and a visualization pipeline to compare runtime and scalability across different parallel models.

PRESENTATIONS

“Increasing IC Model Robustness and Efficiency”

Alexandria, VA

Poster Presentation at the Department of Defense Research Intern Symposium

2024

“Magnetism and Electronic Structure of Ti_4N_3 MXene”

Durham, NC

Oral Presentation at the 67th Electronic Materials Conference at Duke University

2025

“Magnetism and Electronic Structure of Ti_4N_3 MXene”

Raleigh, NC

Poster Presentation at TRIO Ronald E. McNair Research Colloquium

2025

“Training Swarms in Simulated Environments Using Reinforcement Learning”

Alexandria, VA

Oral & Poster Presentation at the Department of Defense Research Intern Symposium

2025

“Electronic and Magnetic Properties of Ti_4N_3 MXene”

Greensboro, NC

Oral Presentation at the DOE-RENEW Symposium at North Carolina A&T State University

2025

“First Principles Study of Ti_4N_3 MXene”

Fayetteville, NC

Oral Presentation at the 2025 Annual Meeting of the North Carolina Academy of Science (NCAS)

2025

“Machine Learning and Machine Learned Predictions in Chest X-ray Images”

Virtual

Oral Presentation at the Association of Nepali Physicists in America (ANPA) Conference

2025

“Neural Networks in Detecting Pneumonia from Chest X-rays”

Atlanta, GA

Accepted Presentation & Travel Award Recipient for Emerging Researchers National (ERN) Conference in STEM

2026

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, SQL, HTML/CSS, R Programming, Haskell

Frameworks: PyTorch, TensorFlow, TorchRL, OpenCV, VASP, ROS, VESTA

Tools: GitHub, Git, Linux, Jupyter Notebook, Adobe Creative Cloud, Excel, Google Collab, CUDA, OpenMP, VMW, AWS

LEADERSHIP & SERVICE

Math Tutor (Fayetteville State University) Current

Providing tutoring for MATH 129, a Pre-Calculus class supporting freshmen, helping them with the order of functions and formulas for lines.

Habitat for Humanity (Fayetteville, NC) 2024

Help with working on drywall and the construction of a new home for a family in need, using primer and paint as well as leveling the floor vinyl.

Peer & Mentor (McNair Scholars Program) Current

Participates in research training, workshops, and mentoring to prepare for graduate-level research and assists peers with research guidance and academic development as part of the cohort

HONORS & AWARDS

Honor Societies: Tau Sigma National Honor Society, Beta Kappa Chi Honor Society, National Society of Leadership and Success, The National Society of Collegiate Scholars

Academic Awards: (2X) Dean's List, (2X) Chancellor's List

Research Awards: NC Space Grant Recipient, MITTIC Award (NASA Innovation Program)

Military Awards: Navy and Marine Corps Achievement Medal, National Defense Service Medal, Global War on Terrorism Service Medal, Navy Pistol Marksmanship Ribbon